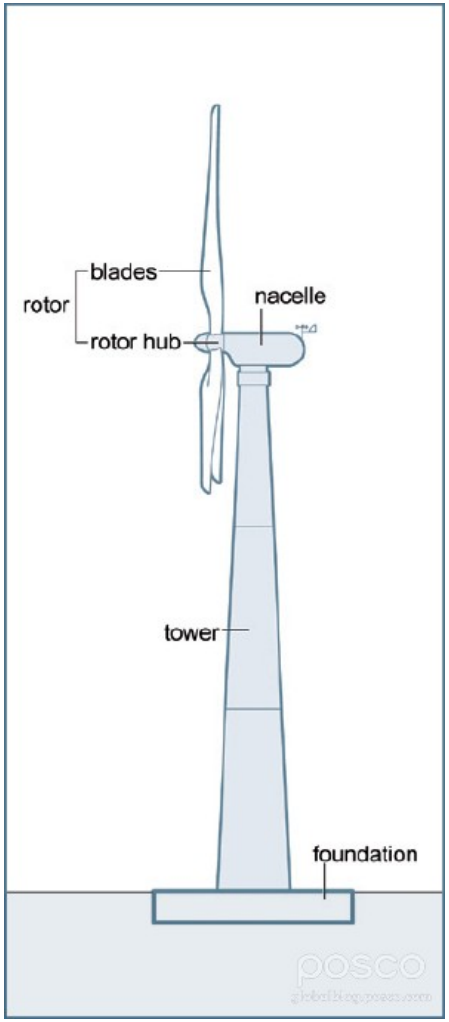
Wind Turbine Model

palm group

Institute of Meteorology and Climatology, Leibniz Universität Hannover

Wind Turbine Model

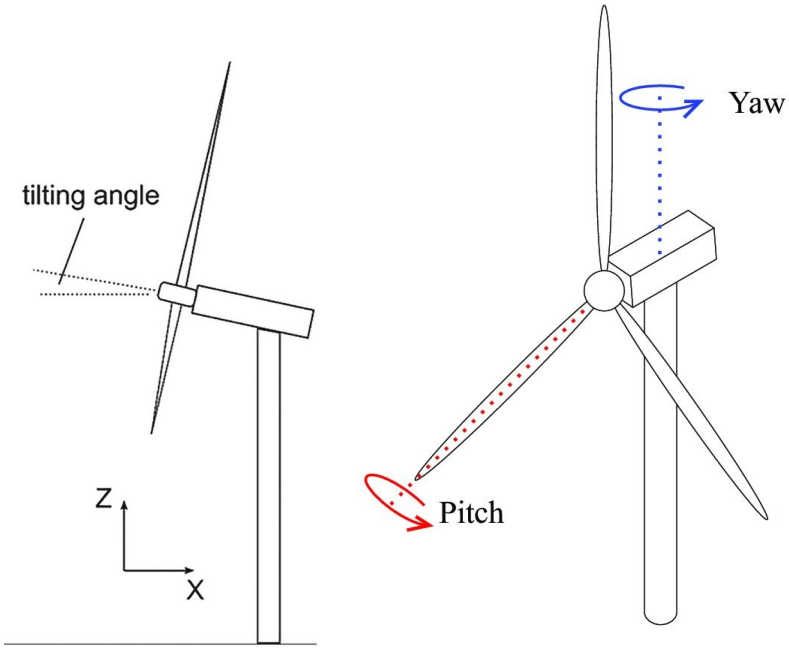
Wind turbine basics



Degrees of freedom

- Yaw angle (horizontal angle of attack to rotor axis)
- Blade Pitch (Rotation of blades to adjust rotation)
- Tilt angle (vertical angle of attach to rotor axis)

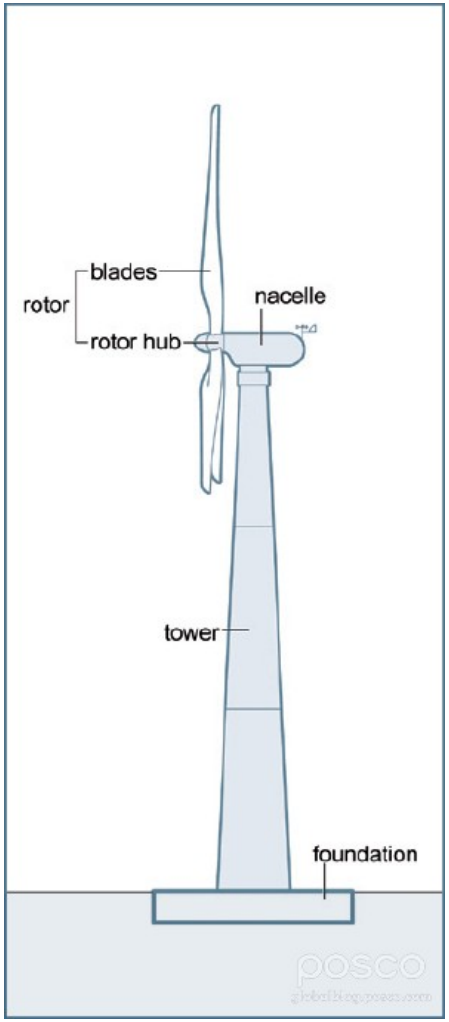
(+ pitch and roll for tower with offshore floating foundations)



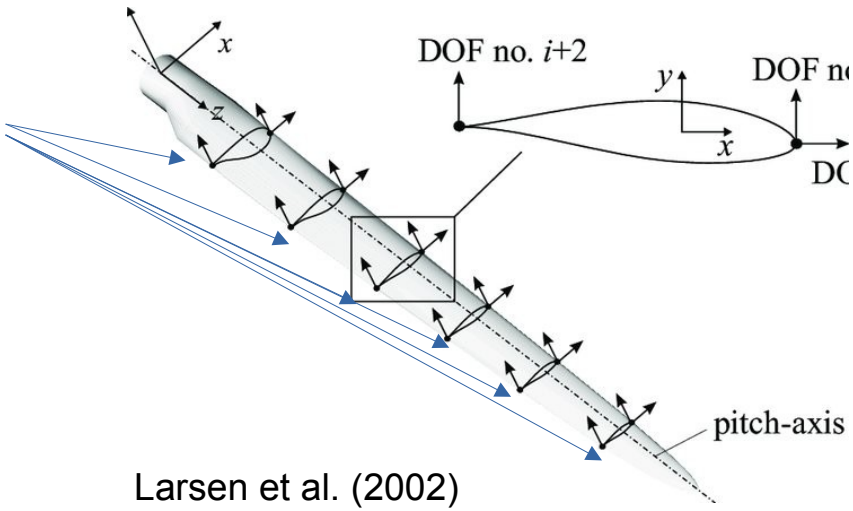
(c) World Steel Association

Wind Turbine Model

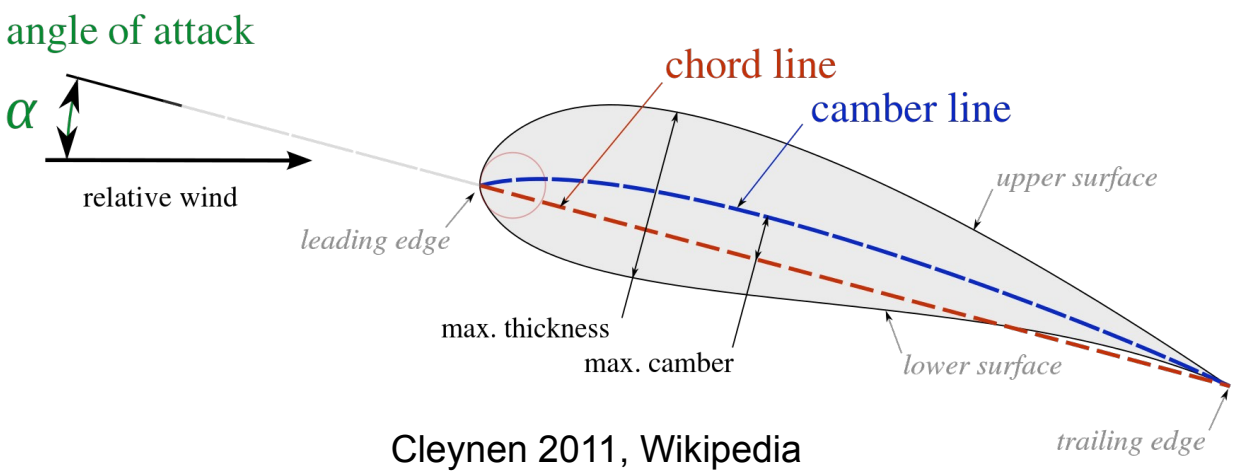
Wind turbine basics



airfoils



Larsen et al. (2002)



Cleynen 2011, Wikipedia

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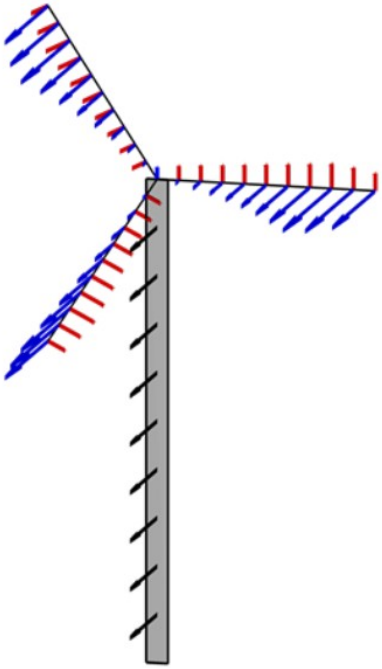
Actuator Line Model (ALM)



- Based on blade element momentum (BEM) method
- Rotors treated as lines, divided into segments
- Lift and drag force are calculated for each segment Δr:

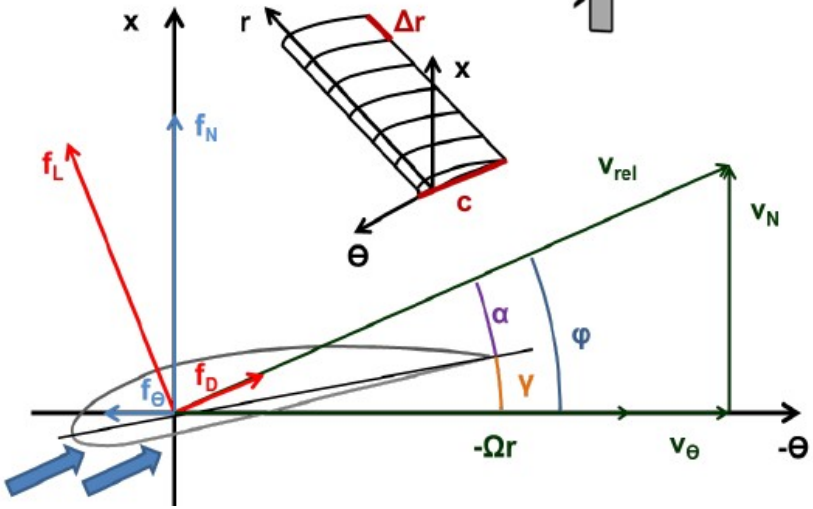
$$F_{L_{b,r}} = \frac{1}{2}\rho C_L V_{rel}^2 c\,\Delta r \qquad F_{D_{b,r}} = \frac{1}{2}\rho C_D V_{rel}^2 c\,\Delta r$$

- Lift C_L and drag C_D coefficients from look-up tables
- Projection on axial (F_N - Torque) and
- tangential (F_θ - Thrust) directions:

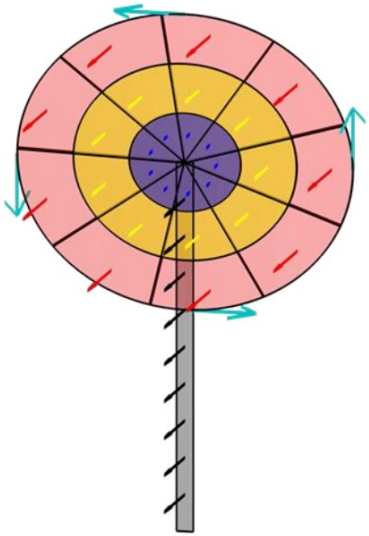
$$F_{\theta_{b,r}} = F_{L_{b,r}} \sin\varphi + F_{D_{b,r}} \cos\varphi$$

$$F_{N_{b,r}} = F_{L_{b,r}} \cos\varphi + F_{D_{b,r}} \sin\varphi$$

$$\varphi = \arctan\left(\frac{V_N}{\Omega r - V_\theta}\right)$$



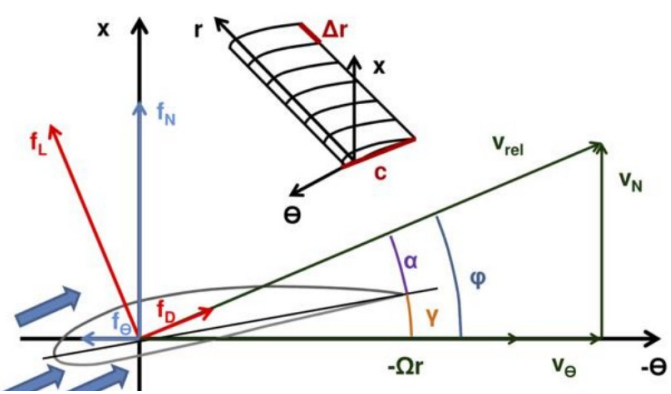
Rotating Actuator Disk Model (ADM-R)



- Rotor disk instead of lines (→ time step issue solved)
- Rotor plane is permeable disk
- Forces are not uniformly distributed (BEM forces)
- (similar to Wu & Porte-Agel, 2011)
- Rotor plane divided into rings and segments
- Lift and drag forces are calculated for each segment:

$$f_l = \frac{1}{2}\rho U_{rel}^2 c_l(\alpha)\frac{N_b c}{2\,\pi r_{seg}}$$ perpendicular to inflow

$$f_d = \frac{1}{2}\rho U_{rel}^2 c_d(\alpha)\frac{N_b c}{2\,\pi r_{seg}}$$ parallel to inflow



(percentage of circumference that is covered with blades)

ρ air density

U_{rel} relative velocity

c_l lift coefficient

c_d drag coefficient

α angle of attack

N_b number of blades

c chord length

r_{seg} radial position of the segment

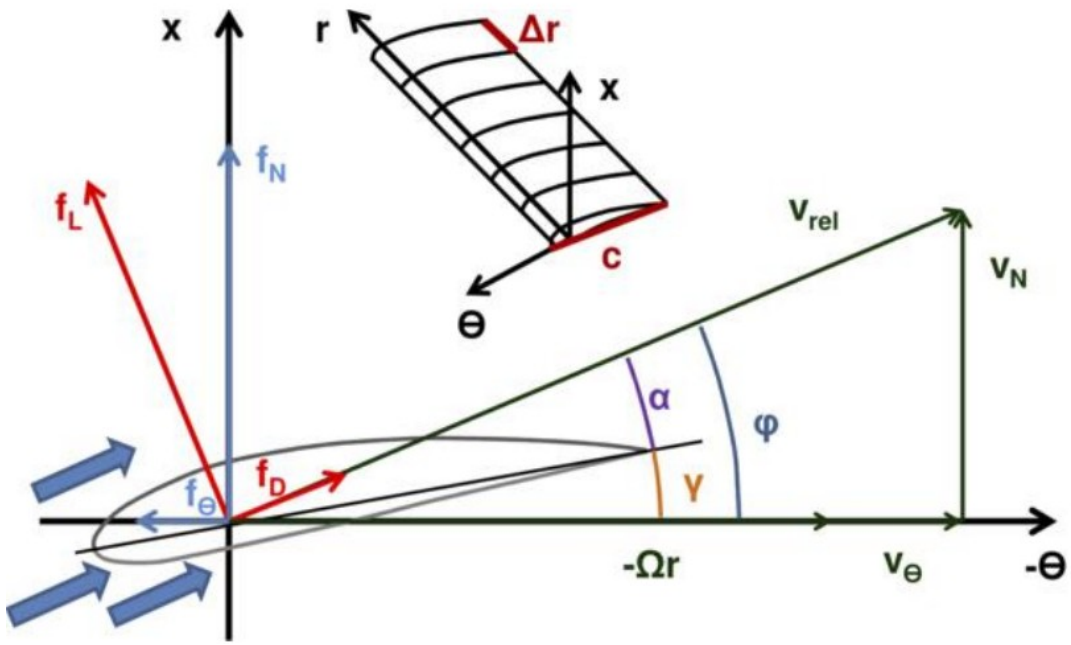
$\frac{N_b c}{2\,\pi r_{seg}}$ solidity factor

Rotating Actuator Disk Model (ADM-R)

- Projection on axial (f_N) and tangential (f_θ) directions as for ALM:

$$f_N = f_l \cos\phi + f_d \sin\phi \qquad f_\theta = f_l \sin\phi - f_d \cos\phi$$

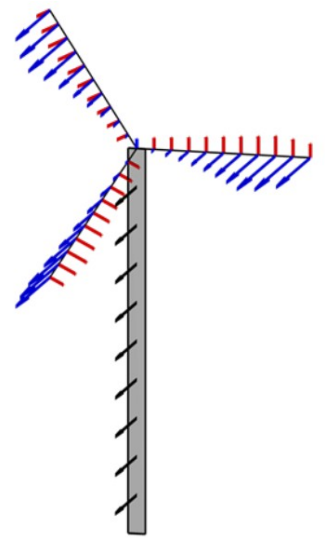
- Smearing to a Gaussian kernel to avoid numerical instabilities

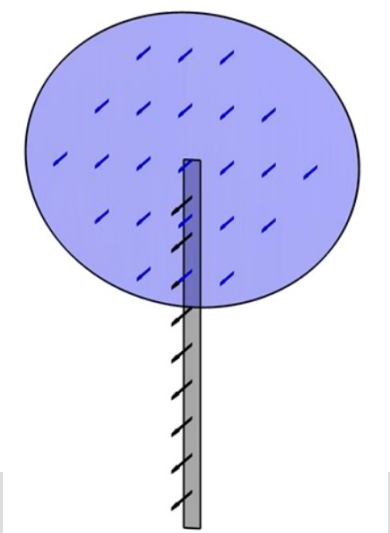


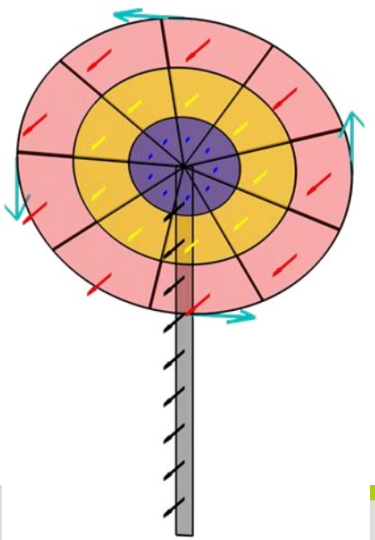
Wind Turbine Model

Overview of typical parameterizations

ALM	ADM	ADM-R
Actuator Line Model (Calaf et al. 2010)	Actuator Disk Model (Troldborg 2008)	Actuator Disk Model with Rotation
Local forces (lines) (BEM equations)	Spatially constant loads	Local forces (segments) (BEM equations)
Rotation (resolved blades)	No rotation	Rotation (blades not resolved)
Very slow (small time step)	Very fast	Fast (significantly faster than ALM)







Wind Turbine Model

Overview of typical parameterizations

- Drag force approach:

$$F_{tower} = -\frac{1}{2}\rho\, C_{D_{tower}}\, V_{ref}^2 \qquad V_{ref} = V_{local} \qquad C_{D_{tower}} = 1.2$$

$$F_{nacelle} = -\frac{1}{2}\rho\, C_{D_{nacelle}}\, V_{ref}^2 \qquad V_{ref} = V_{rotor} \qquad C_{D_{nacelle}} = 0.85$$

- Forces are added to the axial component of the forces generated by the rotor

Wind Turbine Model

How to use the WTM

- Add NAMELIST **&wind_turbine_parameters**

```
&wind_turbine_parameters
         ...
         n_turbines = 2,
         hub_x = 400.0,  400.0,
         hub_y = 500.0, 1500.0,
         ...
/
```

  - For all available parameters refer to:

https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#wind-turbine-parameter

- Add file **<job_identifier>_wtm**, that contains rotor blade data:
  - airfoil, chord and twist distribution along the blade
  - lift and drag coefficients for used airfoils

Wind Turbine Model

How to use the WTM

- Available wind turbines in PALM
  (/palm_model_system_v25.04/packages/palm/model/share/wind_turbines/):
  - NREL 5 MW Reference Offshore turbine
  - IEA 15 MW Reference Offshore turbine

    → _wtm and _p3d example files

- Detailed data about the rotor blades is needed for ADM-R (and ALM), but usually confidential

NREL 5 MW Reference Turbine (Jonkman et al. 2009)

- Representative characteristics for current multi-MW turbines
  - Hub height: 90 m
  - Rotor diameter: 126 m
  - Nacelle diameter: 3 m
  - Tower diameter: 4 m
  - Ω = 12.1 rpm
  - C_L and C_D from look-up tables (depend on angle of attack and distance to hub)
  - Chord and pitch angles from look-up tables

Wind Turbine Model

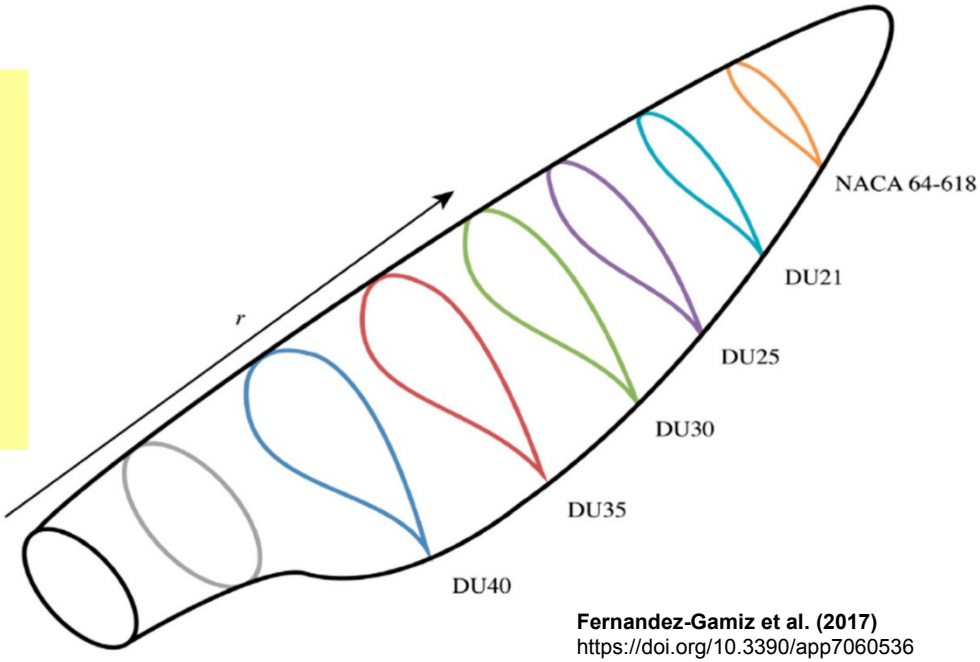
Rotor blade data in _wtm-file (1/2)

- Assign airfoils to radial positions along the blade:

```
# Tablefiles for nrel5MW
# Table 1: Distribution table
# Assigns the airfoils 1-8 to positions along the blade
0.0        1.5       0          0
1.5        5.6       1          1
5.6        8.3       1          2
...        ...
```

- Define twist angle and chord length:

```
# Table 2: Layout file
# Physical layout of the blade
# Pos[m] twist[deg] chord[m]
...        ...        ...
15.9     11.480     4.652
20.0     10.162     4.458
24.1      9.011     4.249
...        ...        ...
```



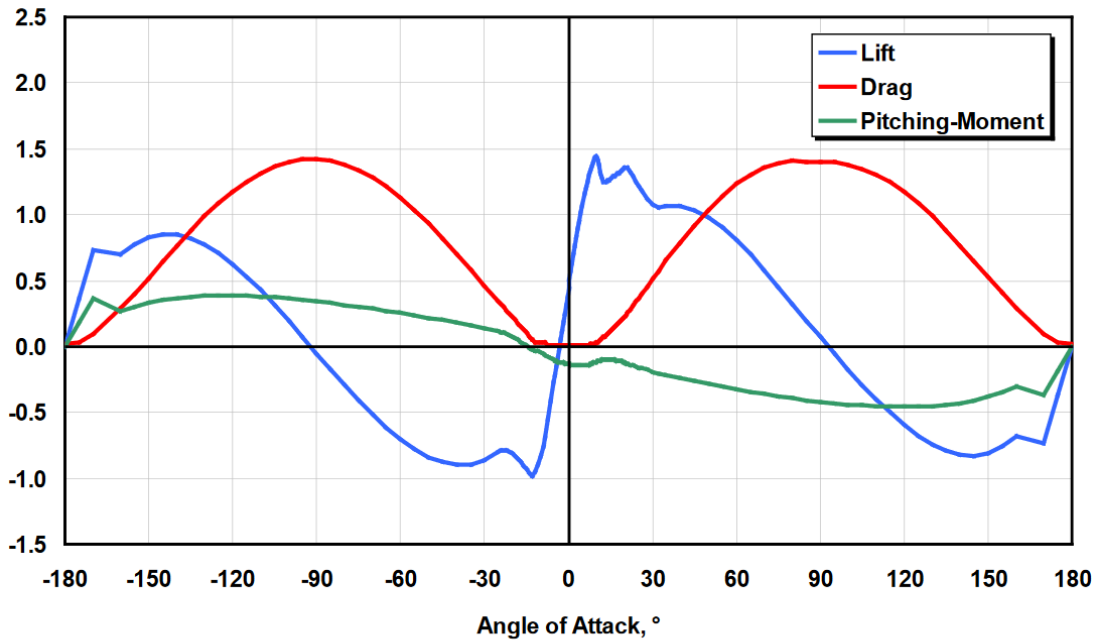
Fernandez-Gamiz et al. (2017)
https://doi.org/10.3390/app7060536

Wind Turbine Model

Rotor blade data in _wtm-file (2/2)

- Define lift and drag coefficients for each airfoil and angle of attack:

```
# Table 3: Lift and drag coefficients for airfoils 1-8
# alpha    ... C_l_3    C_d_3    C_l_4    C_d_4   ...
#
-180.00     ... 0.000   0.0602    0.000   0.0407 ...
-175.00     ... 0.218   0.0699    0.223   0.0507 ...
-170.00     ... 0.397   0.1107    0.405   0.1055 ...
...
```



Corrected coefficients of the DU25 airfoil

Jonkmann et al. (2009) DOI: 10.2172/947422

Wind turbine controler

- Generator torque **speed controller (Jonkman et al. 2009)**
  - Must be adjusted for other turbines
  - Most properties can be adjusted via Fortran NAMELIST (more flexible via NetCDF driver in near future)
- **Pitch controller**
- Active and automatic **yaw controller**
- All controllers can be switched on/off:
  - `speed_control = .F./.T.`
  - `pitch_control = .F./.T.`
  - `yaw_control = .F./.T.`

Wind Turbine Model

Most important parameters

- single parameters, that are valid for all turbines:
  - n_turbines = 2 number of wind turbines
  - pitch_control = .TRUE. activate pitch control
  - yaw_control = .TRUE. activate yaw control

- list parameters, individual for each turbine:
  - hub_x = 400.0, 400.0, x-coordinate of the hub
  - hub_y = 500.0, 1500.0, y-coordinate of the hub
  - hub_z = 90.0, 90.0, z-coordinate of the hub (hub height)
  - rotor_radius = 63.0, 63.0, rotor radius

Wind Turbine Model

Output files

- One output file with time series of operational data of all turbines is saved in /<job_identifier>/OUTPUT/<job_identifier>_wtm.nc
- operational data are e.g.:
  - rotational speed
  - torque
  - power
  - blade pitch angle
  - yaw angle
  - …

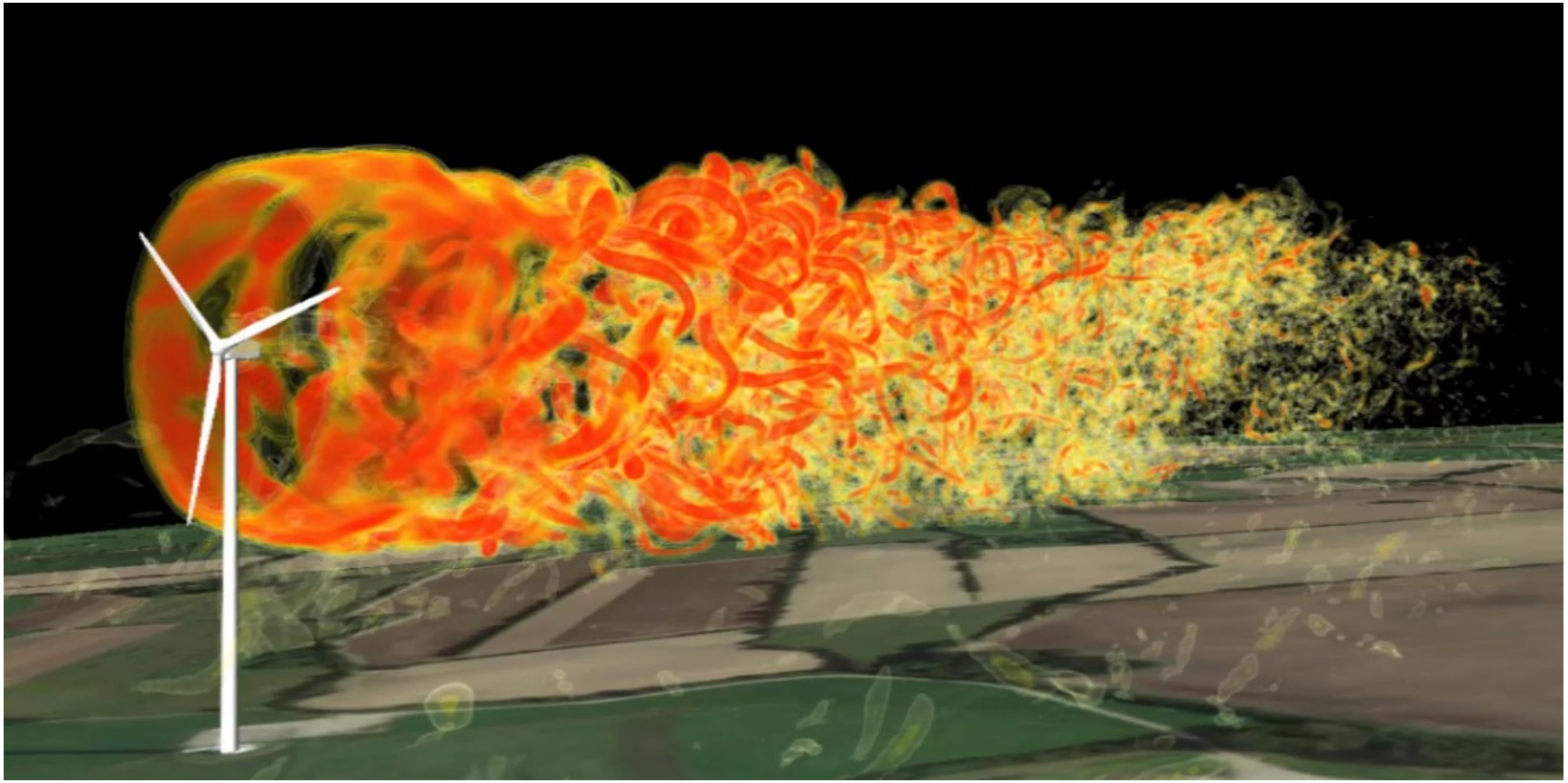
Wind Turbine Model

Suitable setup for wind turbine simulations

- Use non-cyclic boundary conditions:
  - Be aware of the blockage effect when using turbulence recycling!
    - Ensure that the wind turbines have enough distance to the recycling plane
    - 1 – 10 km required, depending on wind farm size and meteorological setup

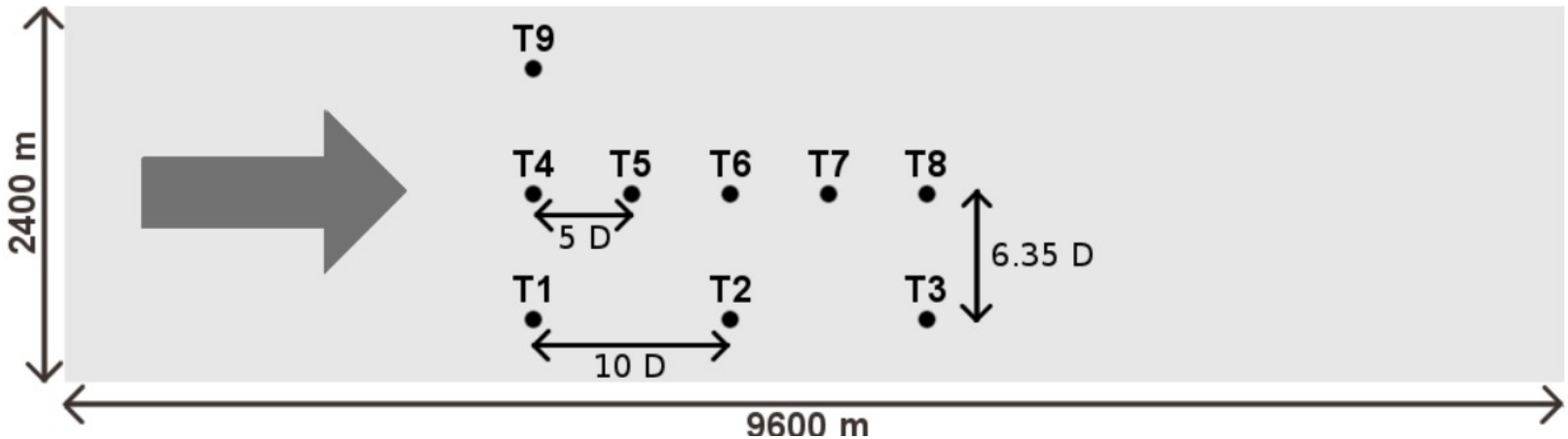
```
&initialization_parameters
        bc_lr                     = 'dirichlet/radiation',
        initializing_actions      = 'cyclic_fill',

        turbulent_inflow          = .TRUE.,
        recycling_width           = 4000.0,
        inflow_damping_height     = 700.0,
/
```

ADM-R in PALM: Visualization



PALM-WTM validation/comparison

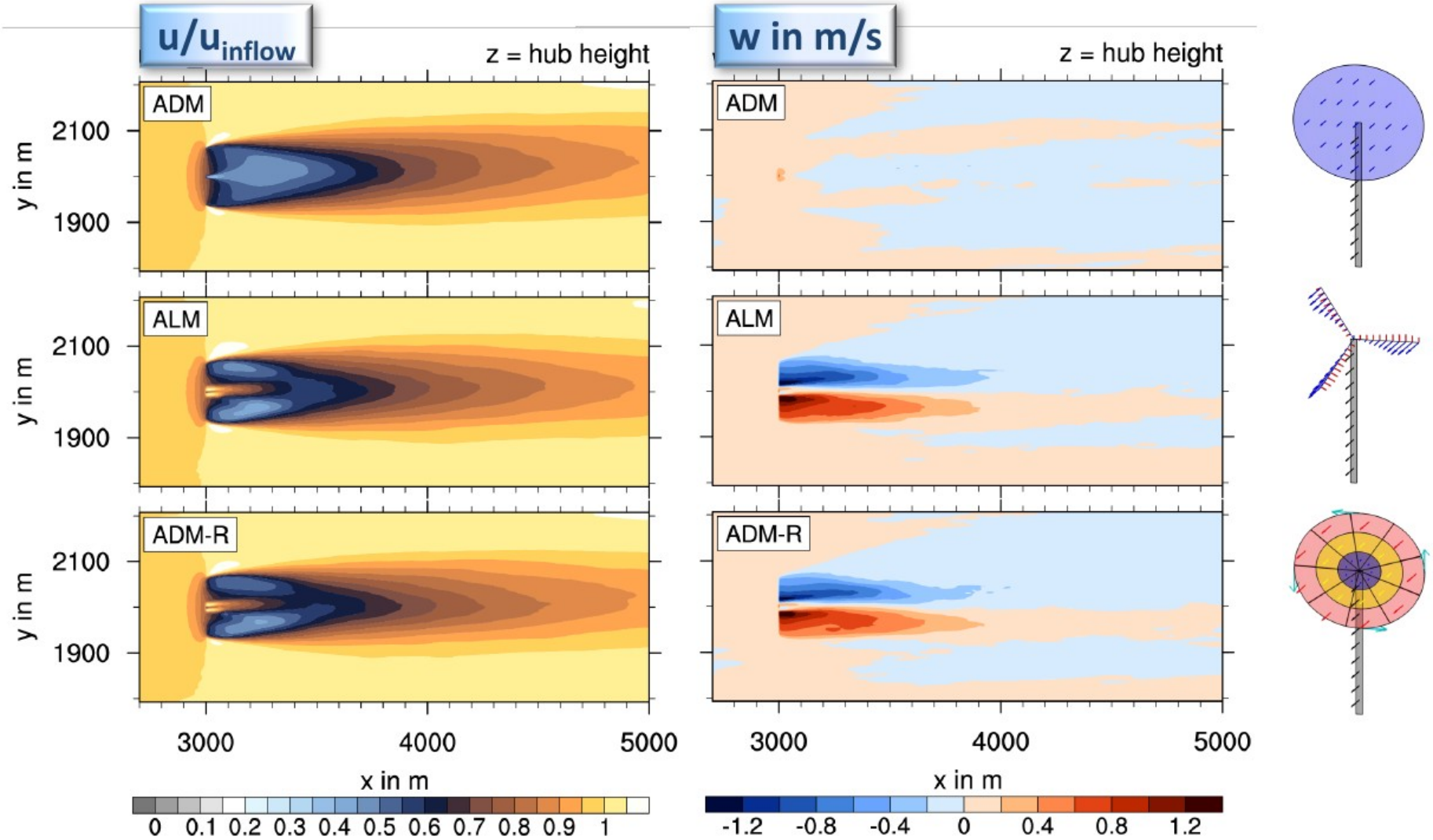
- PALM simulation of an idealized wind farm



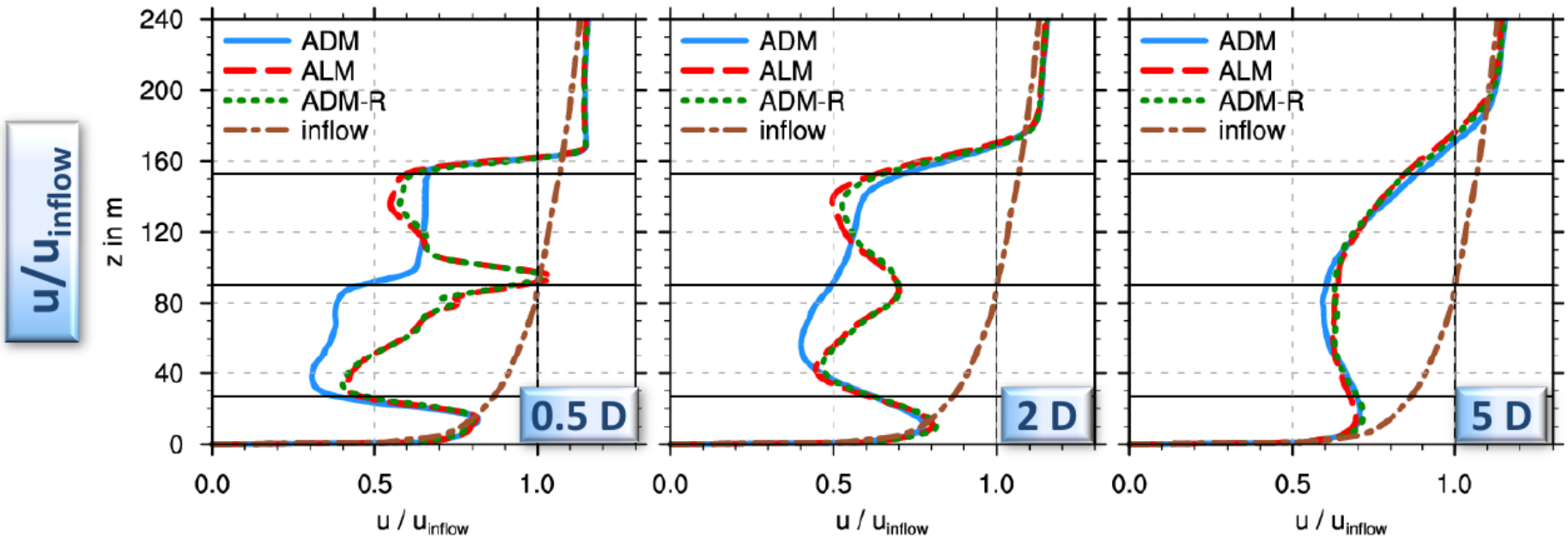
- Offshore conditions
  - z0 = 0.0002 m, $\langle u \rangle_{hub}$ = 8 m/s
  - Neutral statification
- Non-cyclic horizontal boundary conditions with turbulent inflow
- Grid sensitivity test: 16 m, 12 m, 8 m, 4 m, 2 m

Wind Turbine Model

Validation: single wake (T9)



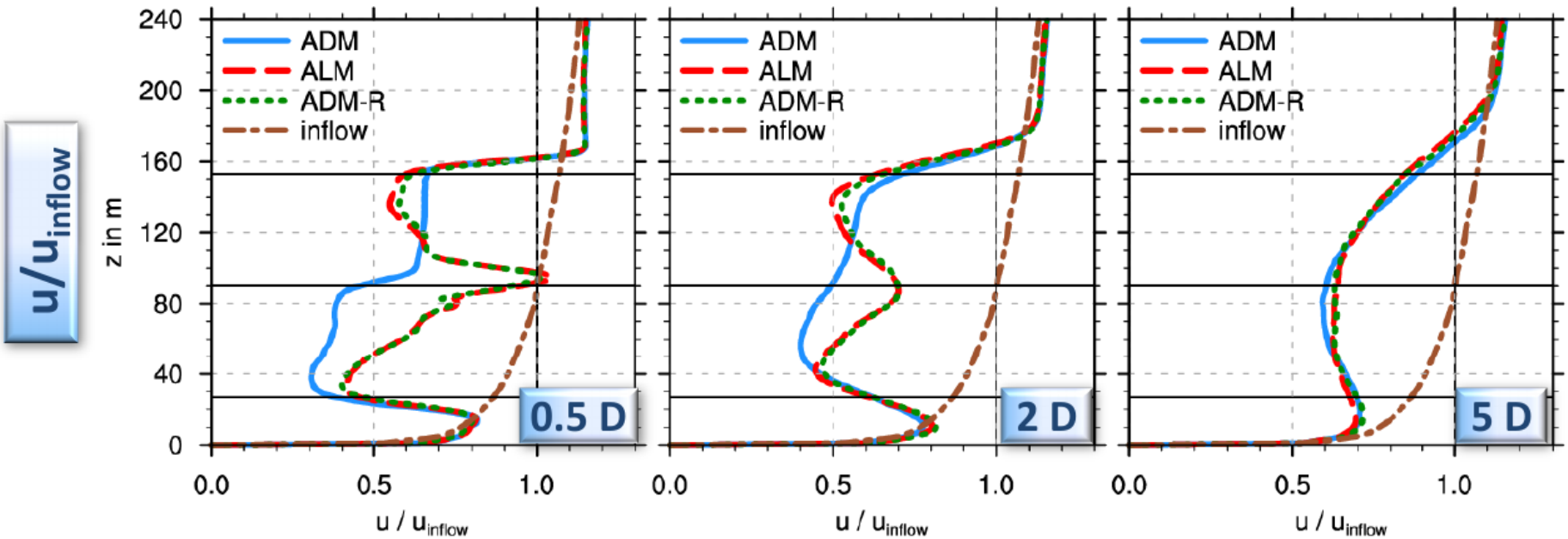
Validation: single wake (T9)



- ALM and ADM-R give almost identical results!
- Near wake is only reproduced ALM and ADM-R
- Far wake is also reproduced by all methods

Wind Turbine Model

Validation: single wake (T9)

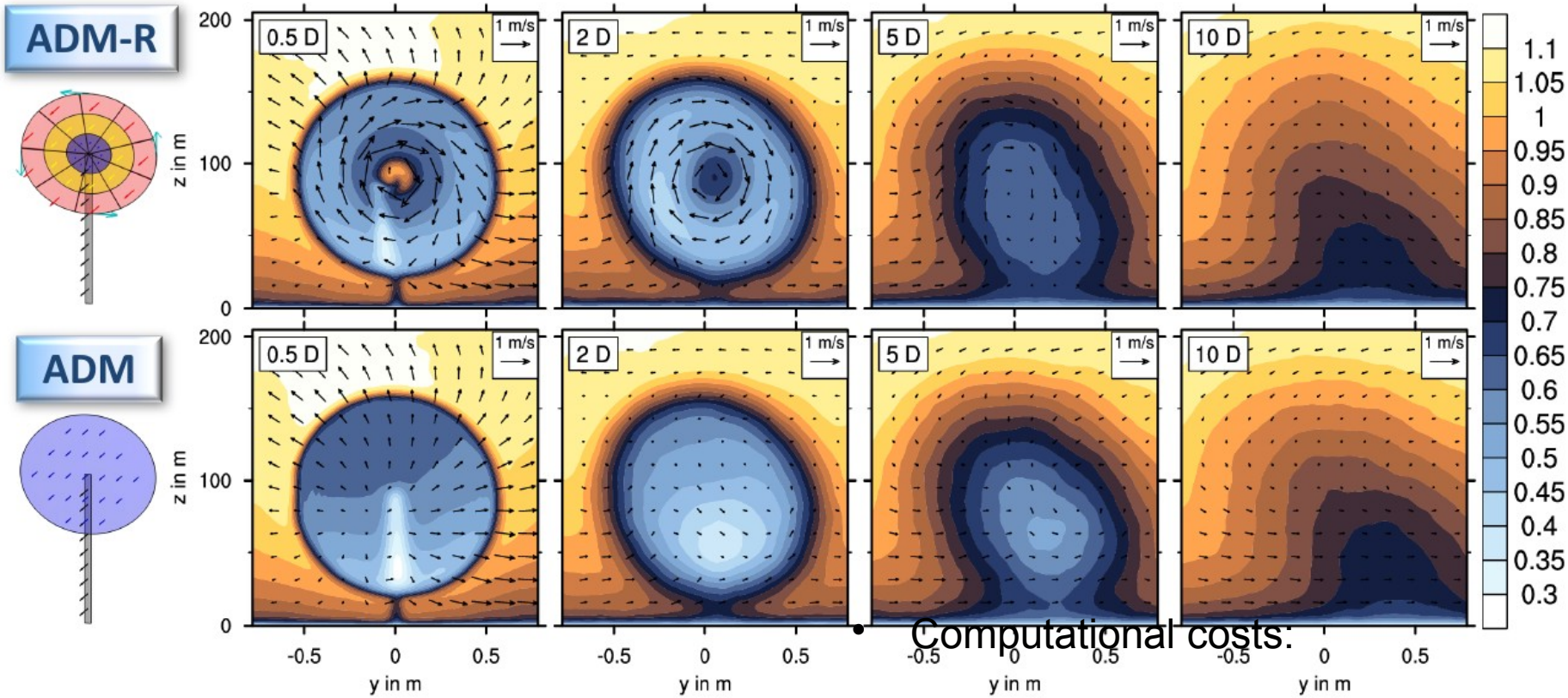


- ALM and ADM-R give almost identical results!
- Near wake is only reproduced ALM and ADM-R
- Far wake is also reproduced by all methods

- Computational costs:

	ADM	ALM	ADM-R
Δ = 8 m	1	12.8	1.5
Δ = 4 m	1	8.1	2.0
Δ = 2 m	1	4.1	2.7

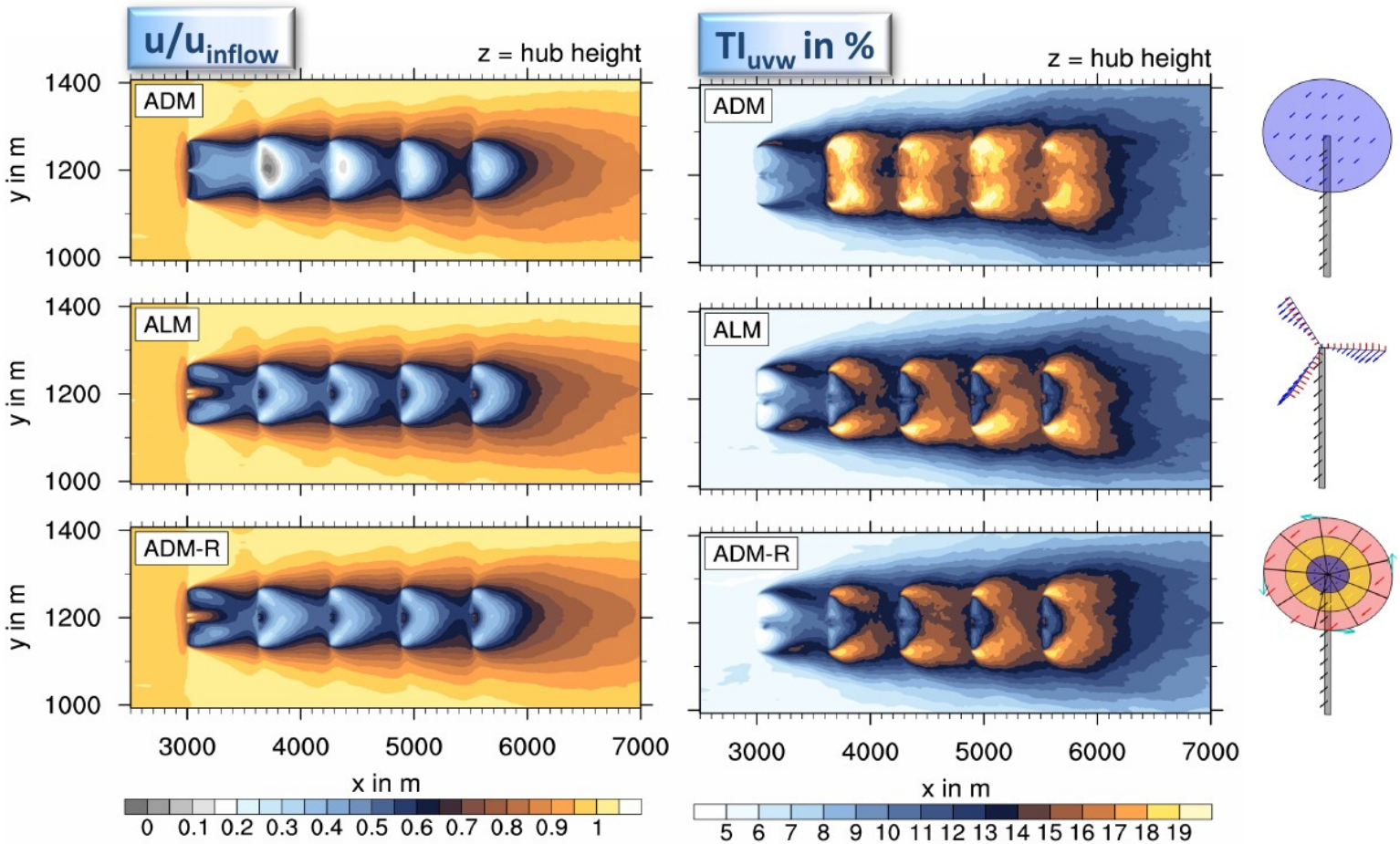
Validation: single wake (T9)



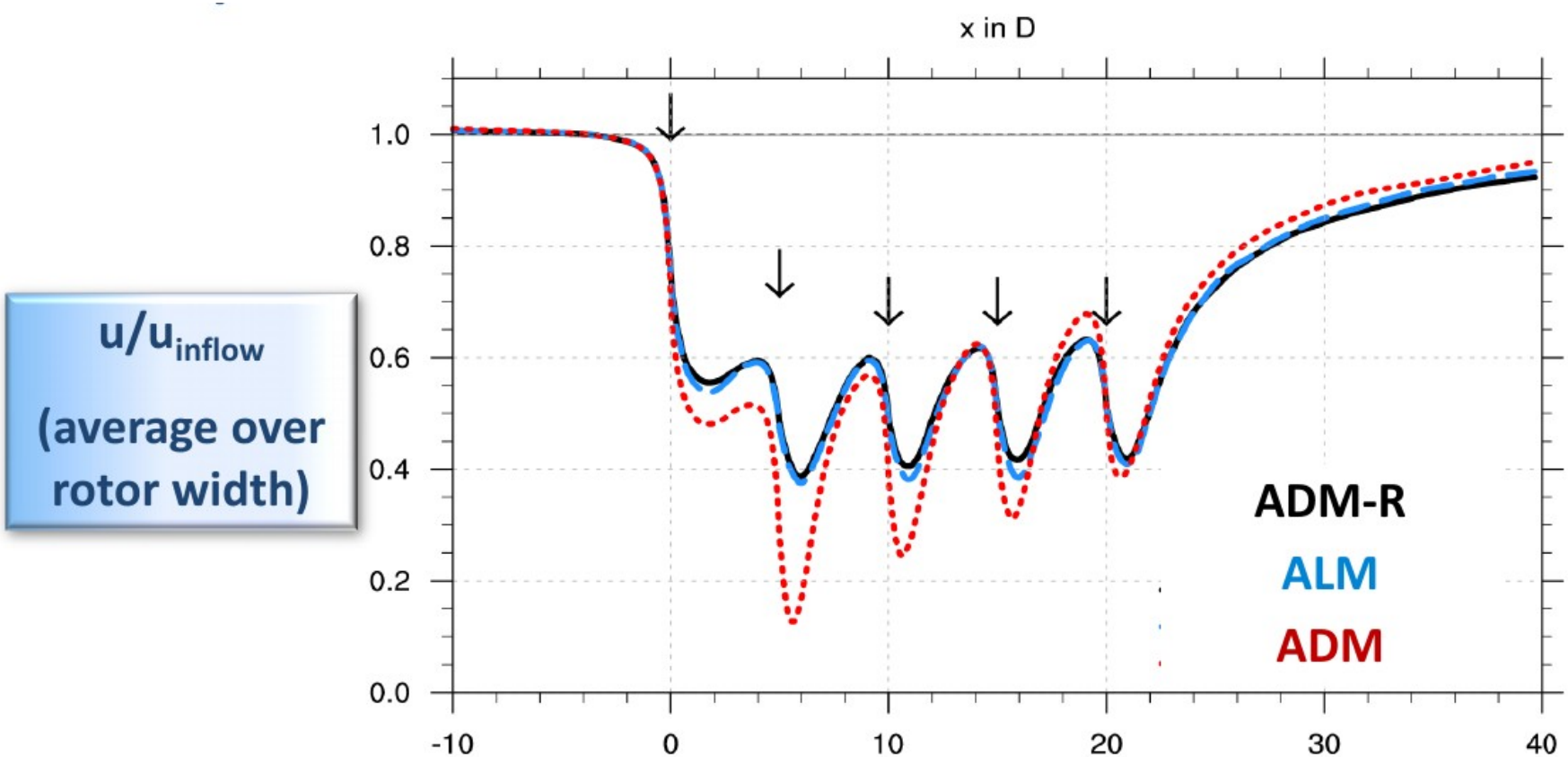
- Computational costs:

- ADM does not capture near wake and wake rotation
- Differences become small in the far wake

Validation: multiple wakes (T4-T8)

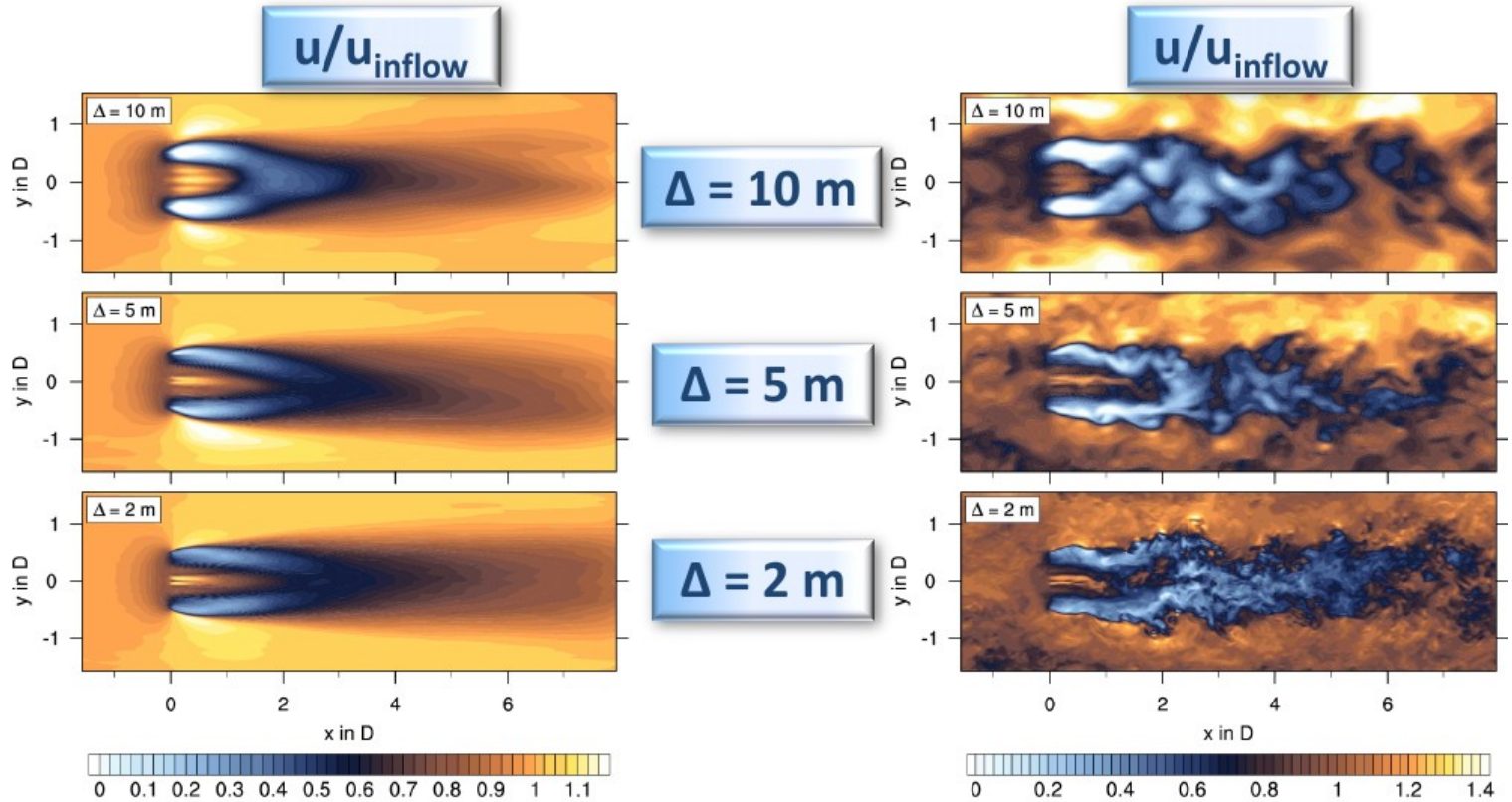


Validation: multiple wakes (T4-T8)



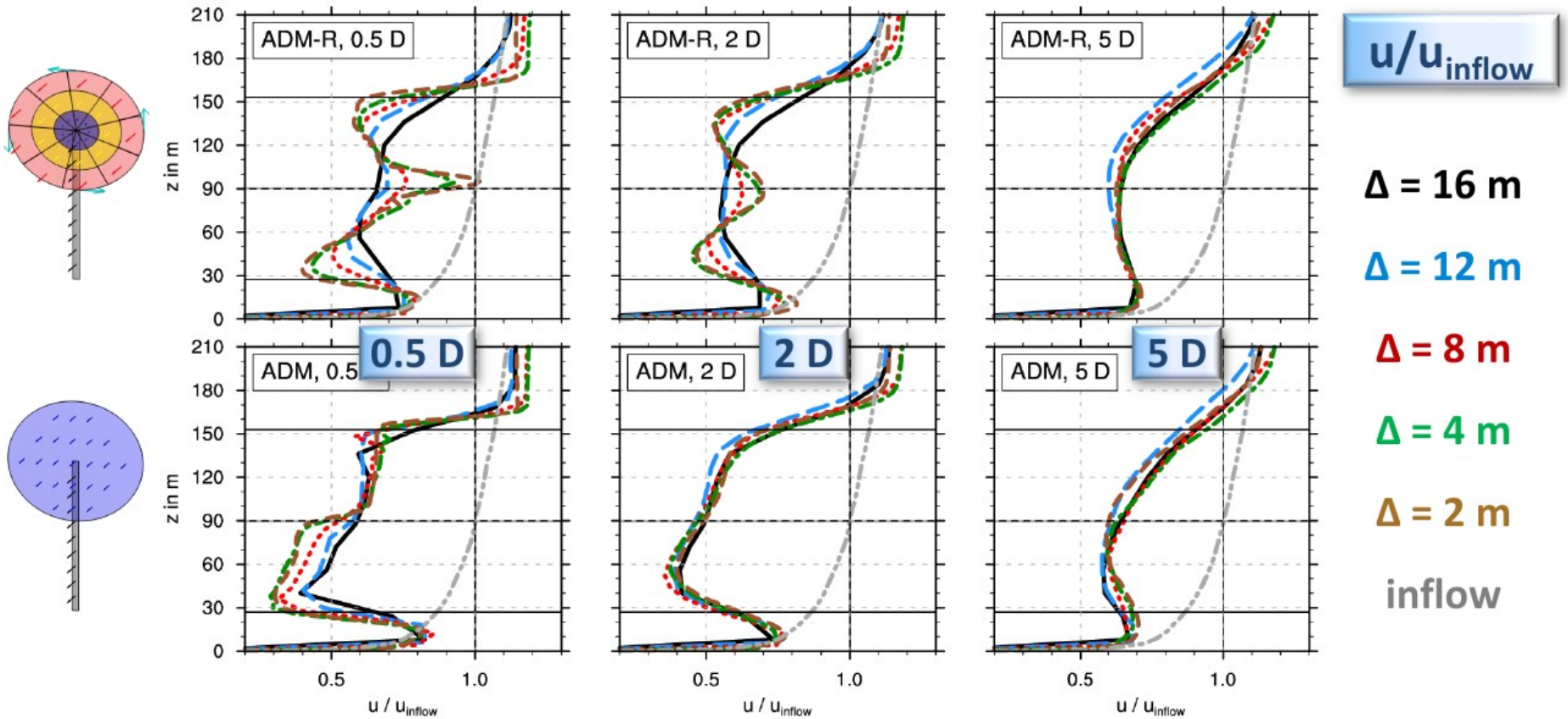
- u-deficit, maximum behind 2nd turbine
- Recovery area due to increased turbulence
- ADM displays strongest deficit and strongest recovery

Validation: grid sensitivity (ADM-R, T9)



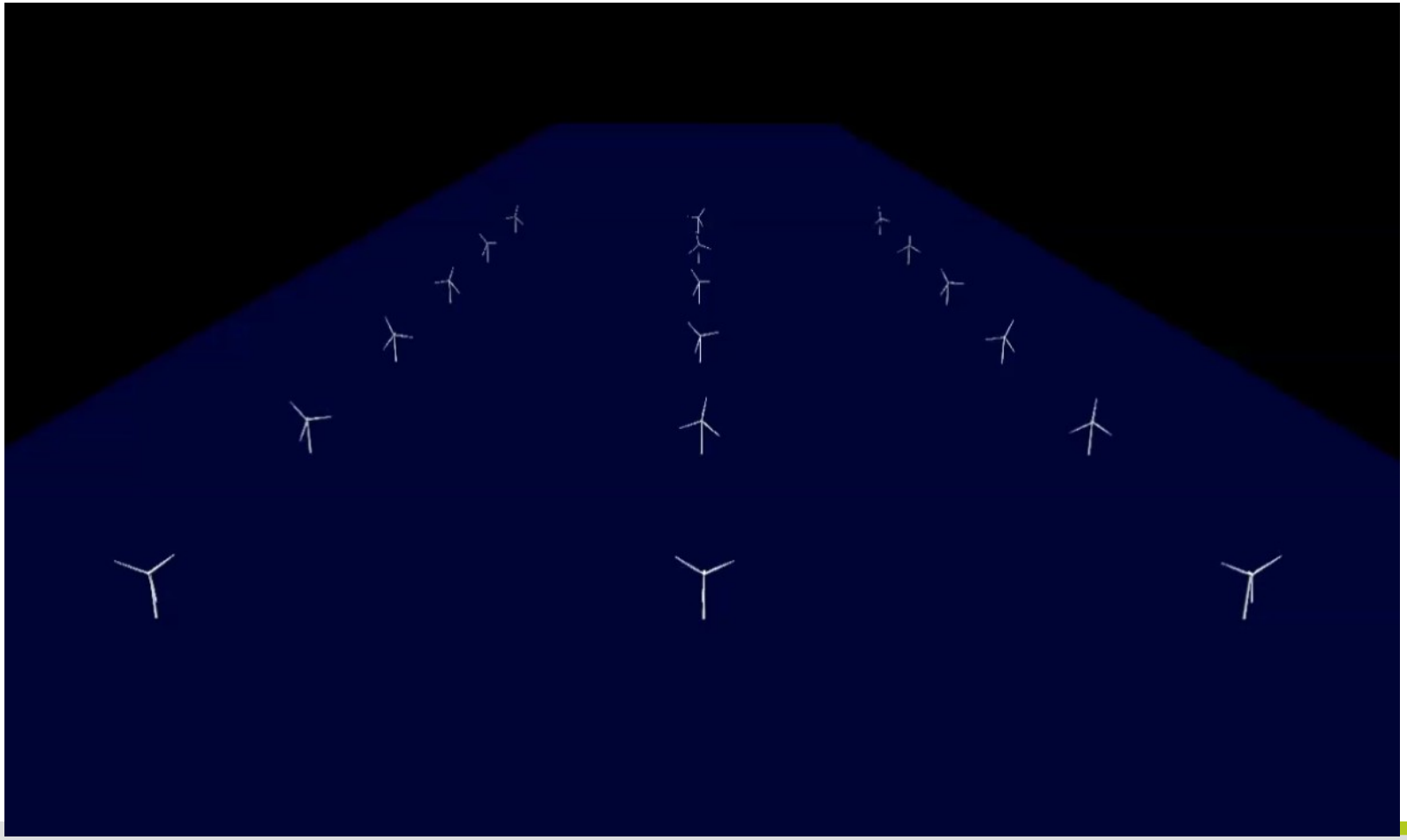
- Large-scale wake structure is well-reproduced at coarse grid
- Finer structures at finer grid spacings
- Longer wake for finer grid spacings

Validation: grid sensitivity (T9)



- Near wake: ADM-R: 30 grid points per rotor required
  ADM: no sensitivity
- Far wake: no sensitivity

Example Application



Wind Turbine Model

Useful URLs

- technical documentation:

https://palm.muk.uni-hannover.de/trac/wiki/doc/tec/wtm

- description of namelist parameters:

https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#wind-turbine-parameters